

2.4 Combinations with Repetition

We have seen that for n distinct objects an arrangement of size r can be obtained in n^r ways, if repetitions are allowed. We now consider the case of the combination of n objects taken r at a time if repetitions are allowed.

Suppose that there are n elements in a set A . We are asked to select r elements from this set, given that each element can be selected multiple times. This is known as a **combination with repetition**.

The number of combinations of n objects taken r at a time with repetition is $(n+r-1)C_r$.

Which is equivalent to the number of integral solutions of the equation

$$x_1 + x_2 + x_3 + \dots + x_n = r, \quad x_i \geq 0, \quad 1 \leq i \leq n.$$

Each non negative solutions of the equation corresponds to a selection, with repetition, of r identical items from n distinct sets are $(n+r-1)C_r$.

Worked Example. 2.4

Example 2.4.1

In how many ways can we distribute 7 bananas and 6 oranges among 4 children so that each child receives at least one banana?

Solution:

After giving one banana to each child, the remaining 3 bananas can be distributed to 4 children with repetition are $(4+3-1)C_3 = 6C_3 = 20$ ways. 6 oranges can be distributed to 4 children in

$(6+4-1)C_4 = 9C_4 = 126$ ways. Therefore, by the rule of product the total number of ways is $20 \times 126 = 2520$.

Example 2.4.2

In how many ways can 10 identical dimes be distributed among 5 children if a) there is no restriction b) each child gets at least one dime.

$$9 + 4 - 1 C_4$$

$$126$$

$$\underline{B} \quad \underline{B} \quad \underline{B} \quad \underline{B} \quad 3+6$$

$$n=4 \quad r=6$$

$$A+6-1 C_6 \quad 9 \times 8 \times 7 \times 6$$

$$56 \times 2 = 112$$

Solution:

Here the children $n = 5$ and dimes $r = 10$.

- a) 10 dimes is distributed to 5 children in $(5 + 10 - 1)C_3 = 14C_{10} = 1001$ ways.
- b) After giving one dime to each child, the remaining 5 dimes is distributed to 5 children in $(5 + 5 - 1)C_5 = 9C_5 = 126$ ways.

Example 2.4.3

In how many ways can a teacher distribute eight chocolates donuts and seven jelly donuts among 3 children helpers if each helper wants at least one donut of each kind.

Solution:

After distributing one chocolates donut to each student, the remaining 5 chocolates donut can be distributed in $(5 + 3 - 1)C_5 = 7C_5 = 21$ ways. For jelly donuts there are $(4 + 3 - 1)C_4 = 6C_4 = 15$ distributions. By the rule of products there are $21 \times 15 = 315$ ways to distribute donuts.

Example 2.4.4

In how many ways can 15 identical candy bars be distributed among five children so that the youngest gets only one or two of them?

Solution:

If the youngest one gets one candy, the number of distributions of 14 candy among 4 children is $(4 + 14 - 1)C_{14} = 17C_{14} = 680$.

If the youngest one gets two candy, the number of distributions of 13 candy among 4 children is $(4 + 13 - 1)C_{13} = 16C_{13} = 560$.

Therefore, by the rule of product the number of distributions $= 680 \times 560 = 38,0800$.

Example 2.4.5

A certain ice cream store has 31 flavors of ice cream available. In how many ways can we order a dozen ice cream cones if (i) we do not want the same flavor more than once? (ii) a flavor may be ordered as many as 12 times? (iii) a flavor may be ordered no more than 11 times?

Solution:

i) 12 ice creams cones can be selected from 31 flavors without repetition is in $31C_{12}$ ways

ii) 12 ice creams cones can be selected from 31 flavors with repetition is in

$$(31 + 12 - 1)C_{12} = 42C_{12} \text{ ways}$$

iii) There are 31 ways to select 12 cones with same flavors. Then a flavor may be ordered no more than 11 times $= 42C_{12} - 31$.

Example 2.4.6

Determine the number of integral solution f $x_1 + x_2 + x_3 + x_4 = 32$

- a) $x_i \geq 0, 1 \leq i \leq 4,$ b) $x_i > 0, 1 \leq i \leq 4,$ c) $x_1, x_2 \geq 5, x_3, x_4 \geq 7,$
 d) $x_i \geq 8, 1 \leq i \leq 4,$ e) $x_i \geq -2, 1 \leq i \leq 4.$

Solution:

a) Each non-negative solution of the equation corresponds to a selection of 32 identical items from 4 distinct collection, repetitions allowed.

The required number of solutions = $(n + r - 1)C_r = (4 + 32 - 1)C_{32} = 35C_{32} = 6545.$

b) $x_i > 0 \Rightarrow x_i \neq 0, 1 \leq i \leq 4,$ then there are $32 - 4 = 28$ identical items. Number of integral solutions are $(4 + 28 - 1)C_{28} = 31C_{28} = 4495.$

c) $x_1, x_2 \geq 5 \Rightarrow x_1, x_2 \neq 0, 1, 2, 3, 4,$ $x_3, x_4 \geq 7 \Rightarrow x_3, x_4 \neq 0, 1, 2, 3, 4, 5, 6$ then there are $32 - 24 = 8$ identical items. $\therefore$ number of integral solutions = $(4 + 8 - 1)C_8 = 11C_8 = 165.$

d) $x_i \geq 8 \Rightarrow x_i \neq 0, 1, 2, 3, 4, 5, 6, 7$ then there are $32 - 32 = 0$ identical items. So number of integral solutions are $(4 + 0 - 1)C_0 = 3C_0 = 1.$

e) putting $y_i = x_i + 2, 1 \leq i \leq 4,$ the equation become

$$y_1 + y_2 + y_3 + y_4 = x_1 + 2 + x_2 + 2 + x_3 + 2 + x_4 + 2 = 32$$

$$y_1 + y_2 + y_3 + y_4 = 40, y_i \geq 0$$

The required number of solutions = $(4 + 40 - 1)C_{40} = 43C_{40} = 12,341.$

Example 2.4.7

Determine the number of integral solutions of $x_1 + x_2 + x_3 + x_4 + x_5 < 40,$ where

- a) $x_i \geq 0, 1 \leq i \leq 5,$ b) $x_i \geq -3, 1 \leq i \leq 4,$

Solution:

We convert the inequality into equality by introducing a new variable $x_6 > 0,$

Then, we get $x_1 + x_2 + x_3 + x_4 + x_5 + x_6 = 40$ where $x_i \geq 0, 1 \leq i \leq 5, x_6 > 0$

Put $x_i = y_i, i = 1, 2, 3, 4, 5,$ and $x_6 - 1 = y_6,$

The equation becomes $y_1 + y_2 + y_3 + y_4 + y_5 + y_6 = 40 - 1 = 39, y_i \geq 0, 1 \leq i \leq 6$

The number of solutions = $(6 + 39 - 1)C_{39} = 44C_{39} = 1,086,008.$

Example 2.4.8

Determine the number of integral solutions of $x_1 + x_2 + x_3 + x_4 + x_5 + x_6 = 10$, where $x_i \geq 0$, $1 \leq i \leq 6$. How many such solutions are there in the inequality

$$x_1 + x_2 + x_3 + x_4 + x_5 + x_6 \leq 10.$$

Solution:

Each non-negative solution of the equation corresponds to a selection of 10 identical balls from 6 distinct sets, repetitions allowed.

The required number of solutions = $(n + r - 1)C_r = (6 + 10 - 1)C_6 = 15C_6 = 350$.

We convert the inequality into equality by introducing a new variable $x_7 > 0$.

Then, we get $x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 = 10$ where $x_i \geq 0$, $1 \leq i \leq 6$, $x_7 > 0$.

Put $x_i = y_i$, $i = 1, 2, 3, 4, 5, 6$ and $x_7 - 1 = y_7$.

The equation becomes $y_1 + y_2 + y_3 + y_4 + y_5 + y_6 + y_7 = 10 - 1 = 9$, $y_i \geq 0$,

$1 \leq i \leq 7$. The number of solutions = $(7 + 9 - 1)C_7 = 15C_7 = 5005$.

Example 2.4.9

Determine the number of integral solutions to the pair of equations

$$x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 = 37, \quad x_1 + x_2 + x_3 = 6, \quad x_i \geq 0, \quad 1 \leq i \leq 7.$$

Solution:

Number of integral solutions of the equation $x_1 + x_2 + x_3 = 6$ is $(6 + 3 - 1)C_3 = 8C_3$.

Number of integral solutions of the equation $x_4 + x_5 + x_6 + x_7 = 37 - 6 = 31$ is $(4 + 31 - 1)C_4 = 34C_4$. By rule of product the number of integral solutions of pair of equations are $8C_3 \times 34C_4$.

Example 2.4.10

Determine the number of integral solutions to the pair of equations

$$x_1 + x_3 + x_5 + x_7 = 5, \quad x_1, x_3, x_5, x_7 > 0 \quad \text{and} \quad x_2 + x_4 + x_6 = 10, \quad x_2, x_4, x_6 > 0$$

Solution:

Since $x_1, x_3, x_5, x_7 > 0 \Rightarrow x_1, x_3, x_5, x_7 \neq 0$, then there are $5 - 4 = 1$ item. Then the number of integral solutions of the equation $x_1 + x_3 + x_5 + x_7 = 1$ is

$$(1 + 4 - 1)C_1 = 4C_1 = 4.$$

Since $x_2, x_4, x_6 > 0 \Rightarrow x_2, x_4, x_6 \neq 0$, then the number of integral solutions of the equation $x_2 + x_4 + x_6 = 10 - 3 = 7$ is $(3 + 7 - 1)C_3 = 9C_3 = 36$. Consequently, by rule of product

number of integral solutions to the pair of equations are $4 \times 36 = 144$.